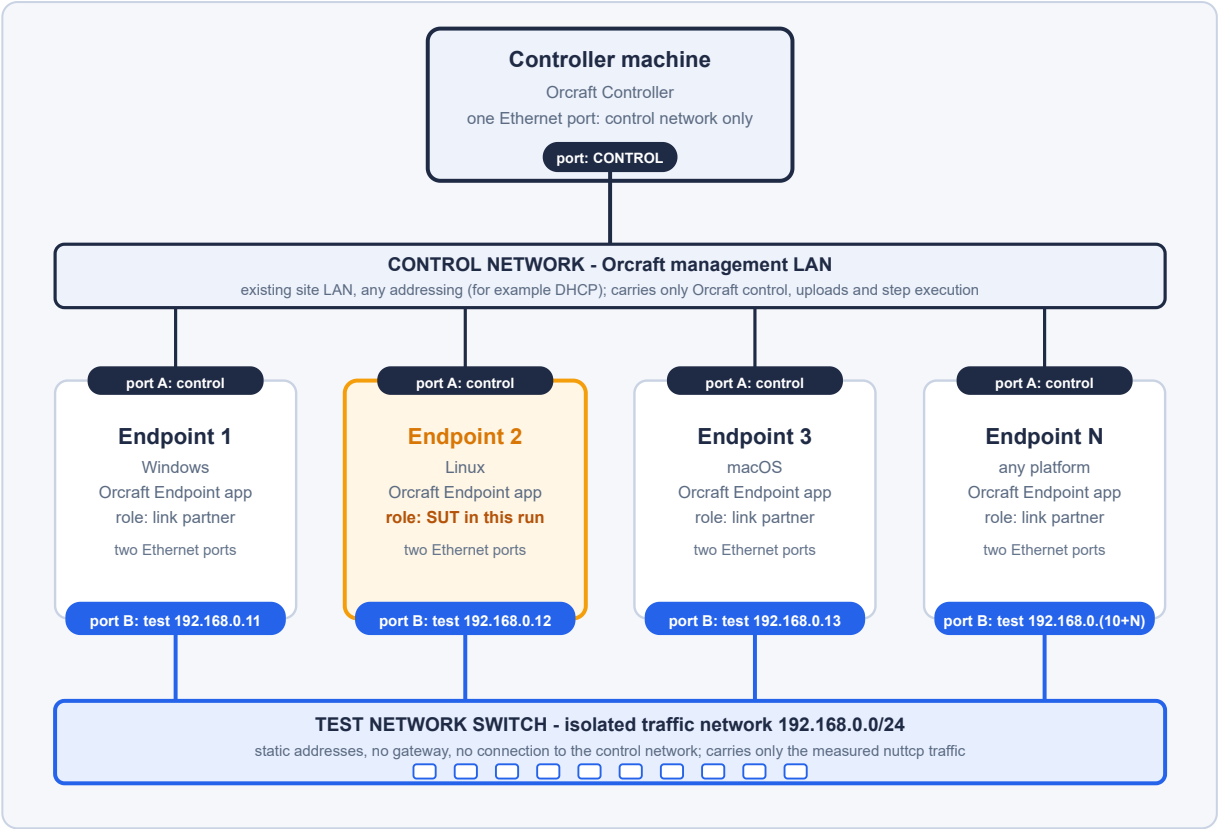
Orcraft Network Test Setup

Two separate networks: an Orcraft control network and an isolated test network

Key facts

- The **controller machine** runs the Orcraft Controller. It has one Ethernet port, on the control network only. It is never a system under test and never a link partner.
- Every **endpoint machine** runs the Orcraft Endpoint app and has **two Ethernet ports**: port A on the control network, port B on the test network.
- The **control network** is the site LAN. The controller reaches every endpoint over it: registration, package upload, step execution .
- The **test network** is a separate Ethernet switch with its own subnet `192.168.0.0/24`. Only the endpoints' test ports are plugged into it. All measured traffic flows here and nowhere else.
- The **system under test (SUT)** is the endpoint whose test port is being measured in a run. It is a runtime role: any endpoint can be the SUT, the others are link partners.
- Endpoints run on **different platforms** (Windows, Linux, macOS). The goal is to measure every test port on every platform.



Networks and Addressing

Each endpoint has two addresses; only the test address carries measured traffic

Addressing plan

Machine	Platform	Port A: control	Port B: test address	Role example
Controller machine	any	site LAN	none	controller
Endpoint 1	Windows	site LAN	`192.168.0.11`	link partner
Endpoint 2	Linux	site LAN	`192.168.0.12`	SUT in this run
Endpoint 3	macOS	site LAN	`192.168.0.13`	link partner
Endpoint N	any platform	site LAN	`192.168.0.(10+N)`	link partner

Address rule: endpoint *k* has test address `192.168.0.(10+k)`, netmask `255.255.255.0`, no gateway. Addresses are static and survive a reboot of the machine. The controller machine has no test address.

Port A is the port the endpoint is registered with in the Orcraft Controller (hostname or LAN address). **Port B** is plugged into the test switch only.

Two addresses per endpoint



Rules for the test suite

- The address Orcraft knows for an endpoint is its **control address**. It must never be used as a nuttcp server or client address.
- Every traffic step must use **test addresses**: a nuttcp server binds on the receiver's test port, a nuttcp client dials the receiver's test address `192.168.0.x`.
- The test address is **discovered on the endpoint itself**: a step running on that endpoint lists its local IPv4 addresses and keeps the one inside `192.168.0.0/24`. The step fails if no such address exists.
- Control traffic and measured traffic never share a port or a cable, so Orcraft control does not disturb the measurement and the measurement cannot break control.

Roles and Test Objective

Every test port on every platform takes its turn as the system under test

Objective

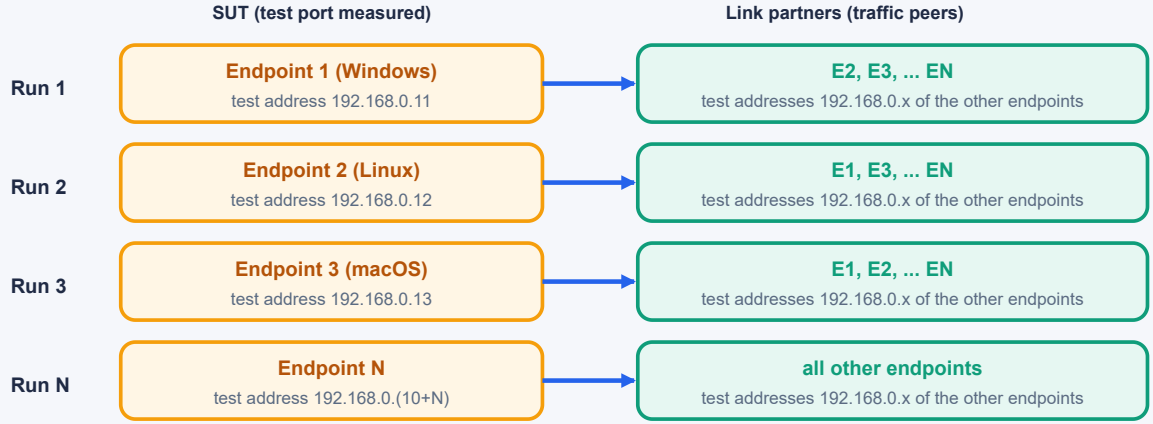
The suite measures the network performance of one machine's **test port** (port B, on the test network). That machine is the **SUT**. All other selected endpoints are **link partners** that send or receive traffic against the SUT over the test switch.

Because the endpoints run on different platforms, the same test set is executed once per endpoint as SUT. The result is a per-platform performance picture of each test port.

Role assignment per run

- **SUT:** a single endpoint, chosen by the user for the run.
- **Link partners:** one or more endpoints, chosen by the user; usually all remaining endpoints.
- The controller machine is never selectable; it has no test port.
- Roles are only meaningful per run. A machine that is the SUT in one run is a link partner in the next.
- Each nuttcp process runs on the endpoint that owns the role: receivers start servers on their own machine, transmitters start clients on their own machine.

Run plan: rotate the SUT



Machine	Runs	Responsibility in a run
Controller machine	Orcraft Controller	Launches the run, uploads the tool package, executes steps on endpoints, collects results. Control network only.
SUT endpoint	Orcraft Endpoint app	Device under test. Its test port transmits (TX), receives (RX) or does both (BX).
Link partner endpoints	Orcraft Endpoint app	Traffic peers on the test network. Receive what the SUT sends, or send traffic to the SUT, all in parallel.

Traffic Direction Modes

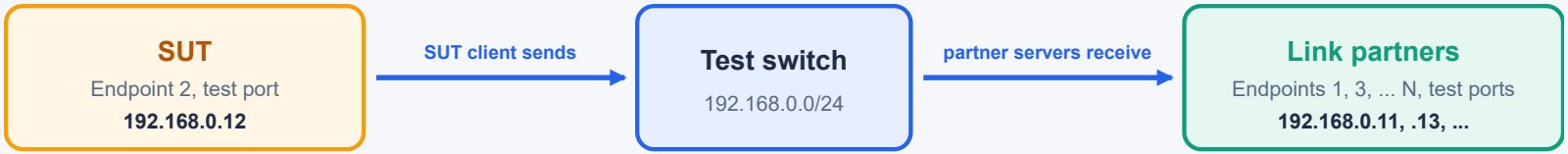
Directions are relative to the SUT; every arrow is on the test network, using test addresses only

Legend

TX = SUT transmits | RX = SUT receives | BX = both directions at the same time, relative to the SUT. nuttcp: the receiver runs the server (`nuttcp -S`), the transmitter runs the client (`nuttcp -t <receiver test address>`). Servers start before clients. With several partners, all partner workers run concurrently.

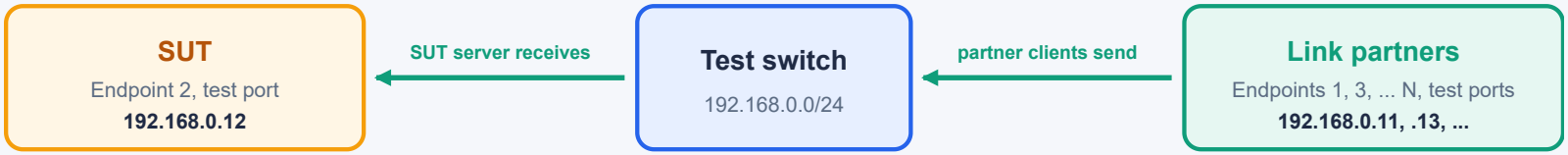
TX test

SUT sends to every link partner. Servers on the partners' test ports, one client per partner on the SUT.



RX test

Every link partner sends to the SUT. Server on the SUT test port, one client on each partner.



BX test

TX and RX at the same time. Servers on both sides, clients on both sides, all workers concurrent.

